

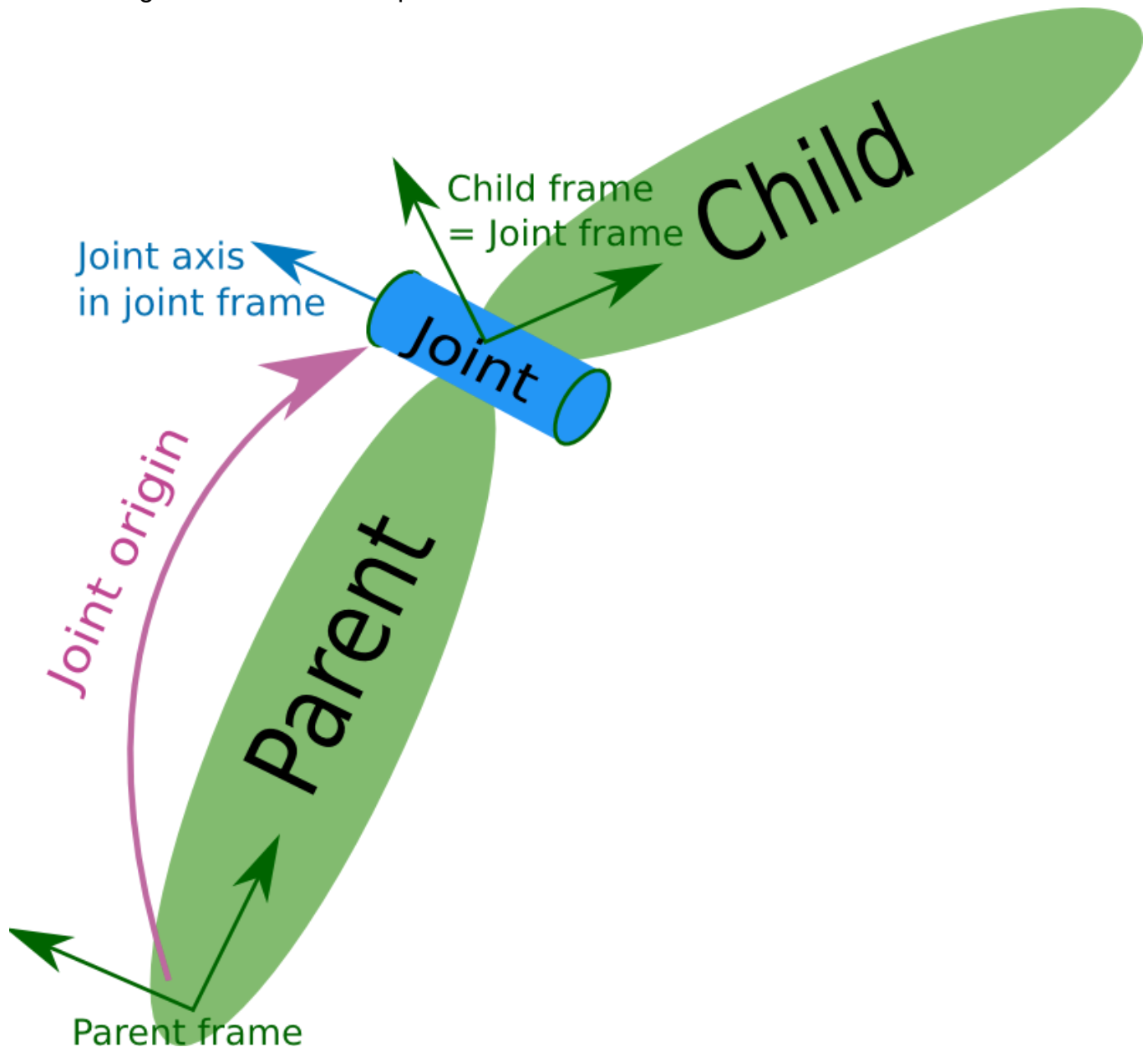
SolidWorks to URDF

Info

Visit wiki.ros.org/urdf/XML for official documentation on the URDF specification

Creating link frames

A joint defines the transform between a parent link and child link. Thus, each link frame is located **at a joint** (aside from the base frame). Create coordinate frames for each link in the model using reference axes and planes.



Creating collision primitives

All primitives must be constructed such that the origin frame of the part is located in the geometric center (i.e., use mid-plane extrusions). Configure all collision geometries as **envelope components** inside assemblies.

Box

To easily generate the dimensions for a collision box on a link, hide the undesired parts and create a bounding box with **Insert** → **Reference Geometry** → **Bounding Box**. Note the dimensions and replicate the box as a separate part. Then, use width mates to center the box on the link.

Cylinder

Ensure the Z axis lies along the height of the cylinder. Lock rotation if mating concentrically.

Finding inertial components

0. Override mass properties of parts/entire link if volume-based calculation is not desired
1. Hide all irrelevant parts
2. **Evaluate** → **Mass Properties**
 - Uncheck *Include hidden bodies/components*
 - *Report coordinate values relative to the link frame*
3. Record the *Center of mass*, this is the xyz offset to apply in the **<inertial>** origin. No rotation will be applied as the moments will be reported aligned to the link frame.
4. Record the *Taken at the center of mass and aligned with the output coordinate system* values, these will be the **<inertial>** values

Important

The default values reported by SolidWorks use a positive tensor notation. **URDF assumes a negative product of inertia convention**. Before inserting the inertia values into the URDF change the setting to *Negative Tensor Notation* in the *Options* of the Mass Properties panel.

Exporting visual meshes

Tip

For assemblies that contain multiple links, use **Display States** to easily separate visual groups.

Single-color links

Each link is exported as an STL. Configure the export settings as:

- Set **Output Coordinate System** to the respective link frame
- Set **Unit** to **Meters**
- Set **Resolution** to **Coarse**
- Check **Do not translate STL output data to a positive space**
- Check **Save all components of an assembly in a single file**

Since STL is purely geometric, the materials must be defined in the URDF.

Multicolor links

Each link is exported as a STEP AP214. Configure the export settings as:

- Set **Output Coordinate System** to the respective link frame
- Check **Export appearances**

Run **meshify.py** on a STEP file or directory of STEP files to generate the associating GLB, DAE, and OBJ meshes.

Note

MuJoCo does not support textured meshes very well. The only approach is to split meshes by material and load each as a separate **<geom>**. For this reason, the OBJ exports are purely geometric, and do not generate a **.mtl** file.